

$\mathbb{N}, \mathbb{W}, \mathbb{I}, \mathbb{Q}, \mathbb{Irrational}, \mathbb{R}.$

$$\emptyset = 0$$

$$\{\emptyset\} = 1$$

$$\{\emptyset, \{\emptyset\}\} = 2.$$

$$\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} = 3$$

$$S_1 = \{x \in \mathbb{Q} : x^2 < 2\}$$

$$S_2 = \{x \in \mathbb{Q} : x^2 > 2\}$$

there is no max. number in S_1

and there is no min. number S_2 .

$$x_1 \in S_1, x_1^2 < 2.$$

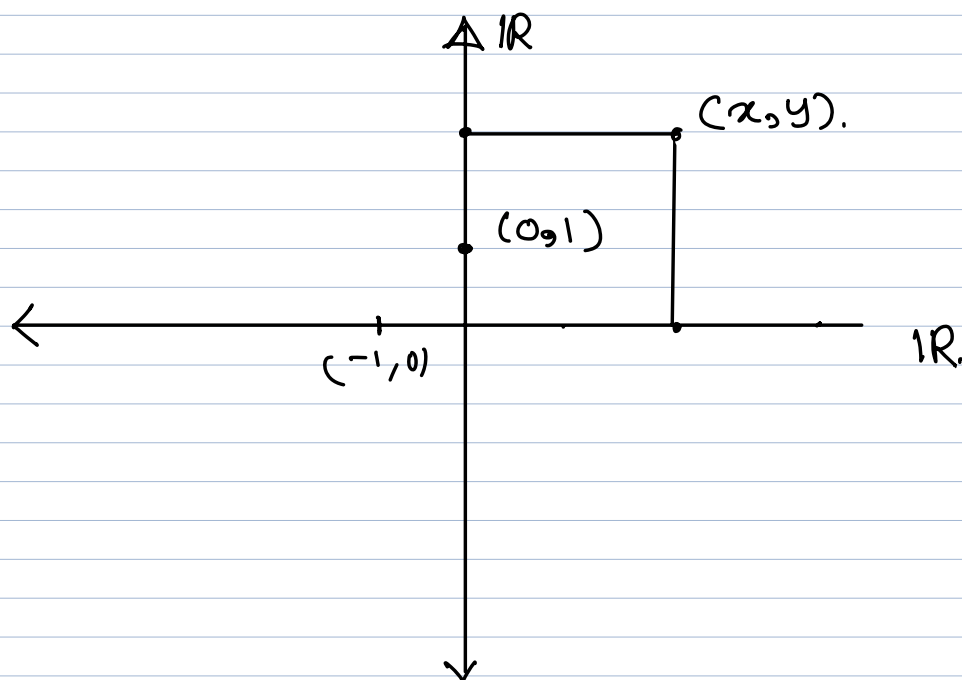
$$\forall x \in S_1 \exists x' \in S_1 \ni x' > x$$

$$\forall x \in S_2 \exists x' \in S_2 \ni x' < x.$$

$$\sqrt{2} = \sup(S_1) = \inf(S_2)$$

$\hookrightarrow$ supremum = least
upper
bound.

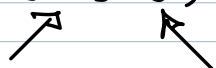
infimum
greatest
lower
bound



Complex Number is an ordered pair of real numbers.

Notation: (x, y) where $x, y \in \mathbb{R}$.

(x, y)



Real Part Imaginary Part

Let $C_1(x_1, y_1)$ and $C_2(x_2, y_2)$ be two Complex numbers.

$$C_1 + C_2 = (x_1 + x_2, y_1 + y_2)$$

$$C_1 - C_2 = (x_1 - x_2, y_1 - y_2)$$

$$C_1 \times C_2 = (x_1 \cdot x_2 - y_1 \cdot y_2, x_1 \cdot y_2 + x_2 \cdot y_1)$$

$$C_1 / C_2 = \left(\frac{x_1 \cdot x_2 + y_1 \cdot y_2}{x_2^2 + y_2^2}, \frac{y_1 \cdot x_2 - x_1 \cdot y_2}{x_2^2 + y_2^2} \right)$$

$$\boxed{i = (0, 1)}$$

$$i \times i = \overset{x^1}{(0, 1)} \times \overset{x^2}{(0, 1)}$$

$$= (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 0 \cdot 1)$$

$$= (0 - 1, 0 + 0)$$

$$= (-1, 0)$$

$$i^2 = -1$$

$$\boxed{i = \sqrt{-1}}$$

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta) = e^{in\theta}$$

$$\boxed{e^{i\pi} + 1 = 0}$$

$$C = (x, y)$$

$$n \in \mathbb{R}$$

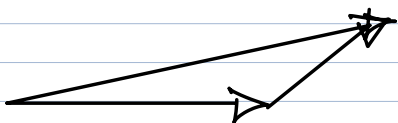
$$n \cdot C = (n \cdot x, n \cdot y)$$

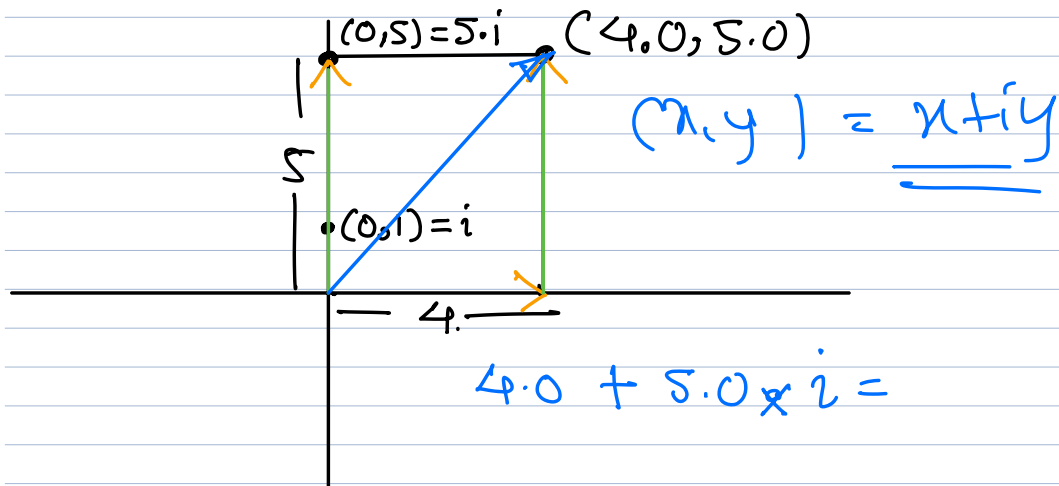
$$C = (1.1, 2.2)$$

$$5 \cdot C = 5 \cdot (1.1, 2.2)$$

$$= (5, 0) \circ (1.1, 2.2)$$

$$= (5 \times 1.1, 5 \times 2.2)$$



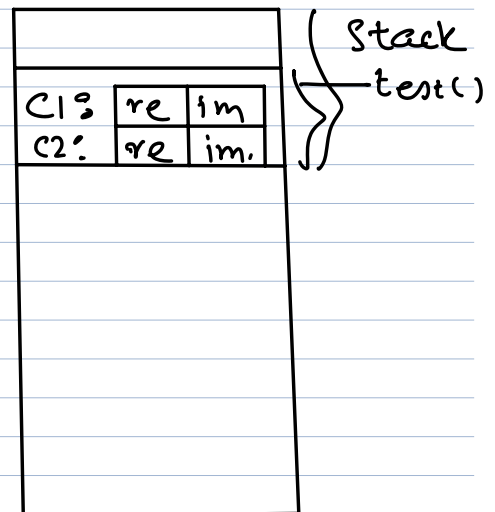


```

Void test (Void)
{
    complex c1 (1.1, 2.2);
    complex c2 (3.3, 4.4);

    //code
    complex sum = c1.add(c2);
}

```



$x + y$ $y + x$ $+, *$

$x * y$ $y * x$

$x - y$ $y - x$

x / y y / x

Complex::add(&c1, c2);
 → complex type object by value return.

CALLER: `c1.add(c2)`

`Complex::add(&c1, c2)`

DEFINITION WRITER:

`class Complex`
`{`
 `public:`

`Complex c2`
`Complex& c2`

`Complex add(const Complex& c2) const`

HIDDEN THIS PTR:

`const Complex* this`

`Complex add(const Complex& c2) const`
`{`

`}`